



Visualization Tool Using t-SNE

Visualizing Data using t-SNE

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Dimensionality reduction $\{x_n\}_{n=1}^N \in \mathbb{R}^D$

$$d \ll D.$$

$d=2$ for visualization.

Neighborhood Embedding

$$\{y_n\}_{n=1}^N \in \mathbb{R}^d$$
$$y_n = \underset{\substack{\text{largest} \\ \text{variance}}}{W^T} (x_n - \mu)$$

(x_i, x_j) are close in D dimensional space

$\rightarrow \{y_i, y_j\}$ to be close in d dimensional space

Dimensionality reduction

↳ preserving neighborhood

probability model

and distance between probability
distr

Visualization tool using t-SNE

$$p_{ii} = 0$$

Gaussian
 $\mathcal{D}^D$

student
t-distribution

$$q_{ii} = 0 \quad \forall i$$

$$p_{ij} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma^2)}{\sum_{k \neq l} \exp(-\|x_k - x_l\|^2 / 2\sigma^2)}$$

width parameter σ

normalization

$\mathbb{R}^D$

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

$\mathbb{R}^D$

$$C = KL(P || Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

KL divergence

$-\sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$

Goal in t-SNE

arg min
 $\{y_1, \dots, y_N\}$

C.
↑

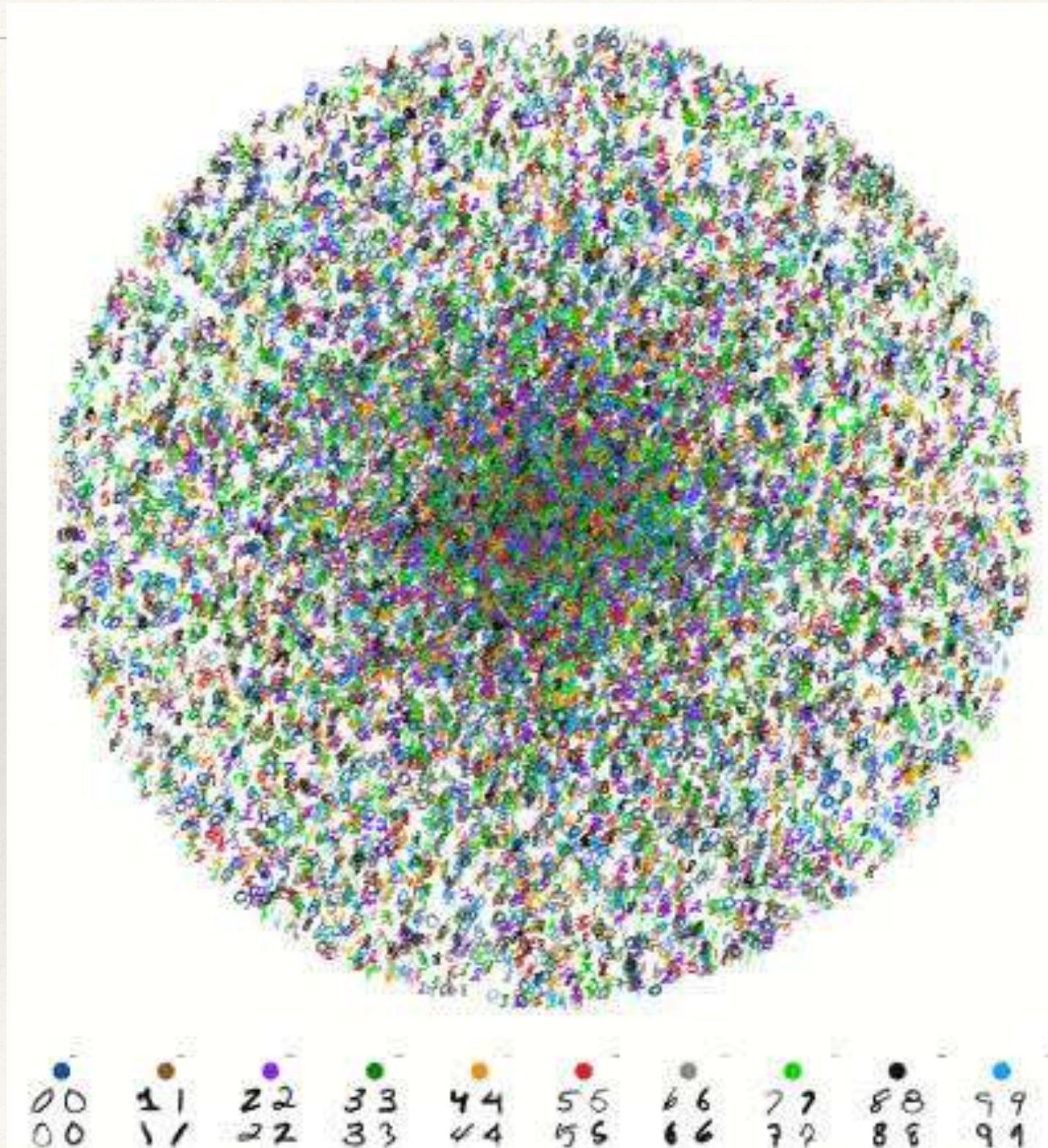
$$= \underline{y_1^*}, y_2^* \dots y_N^*$$

Gradient descent

$$y_i \in \underline{\mathbb{R}^d}$$

Iterative update of lower dimensional representation.

Visualizing back propagation in tSNE



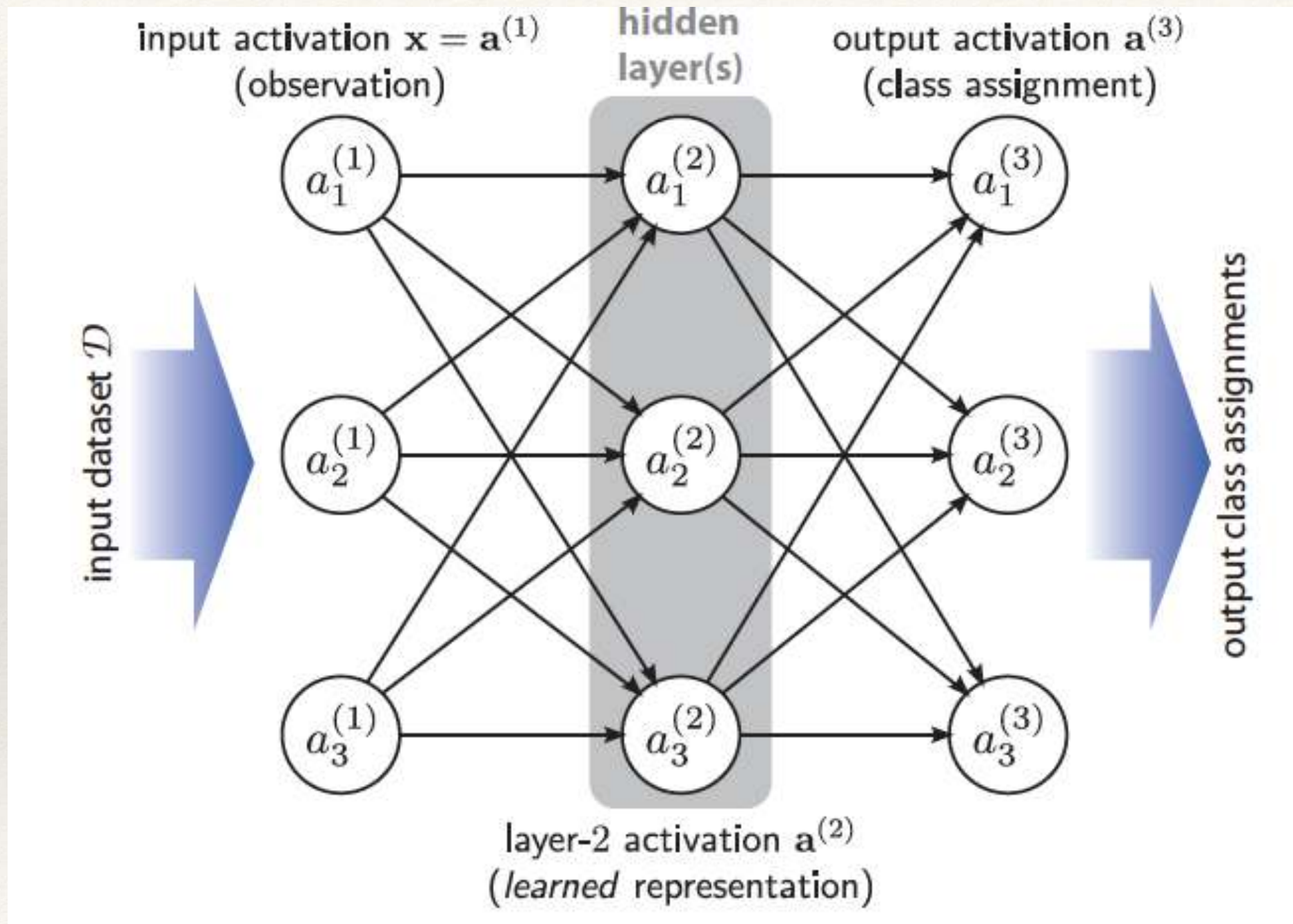
Understanding Deep Networks

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS VOL. 23, NO. 1, JANUARY 2017

Visualizing the Hidden Activity of Artificial Neural Networks

Paulo E. Rauber, Samuel G. Fadel, Alexandre X. Falcão, and Alexandru C. Telea

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SVHN dataset



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CIFAR-10

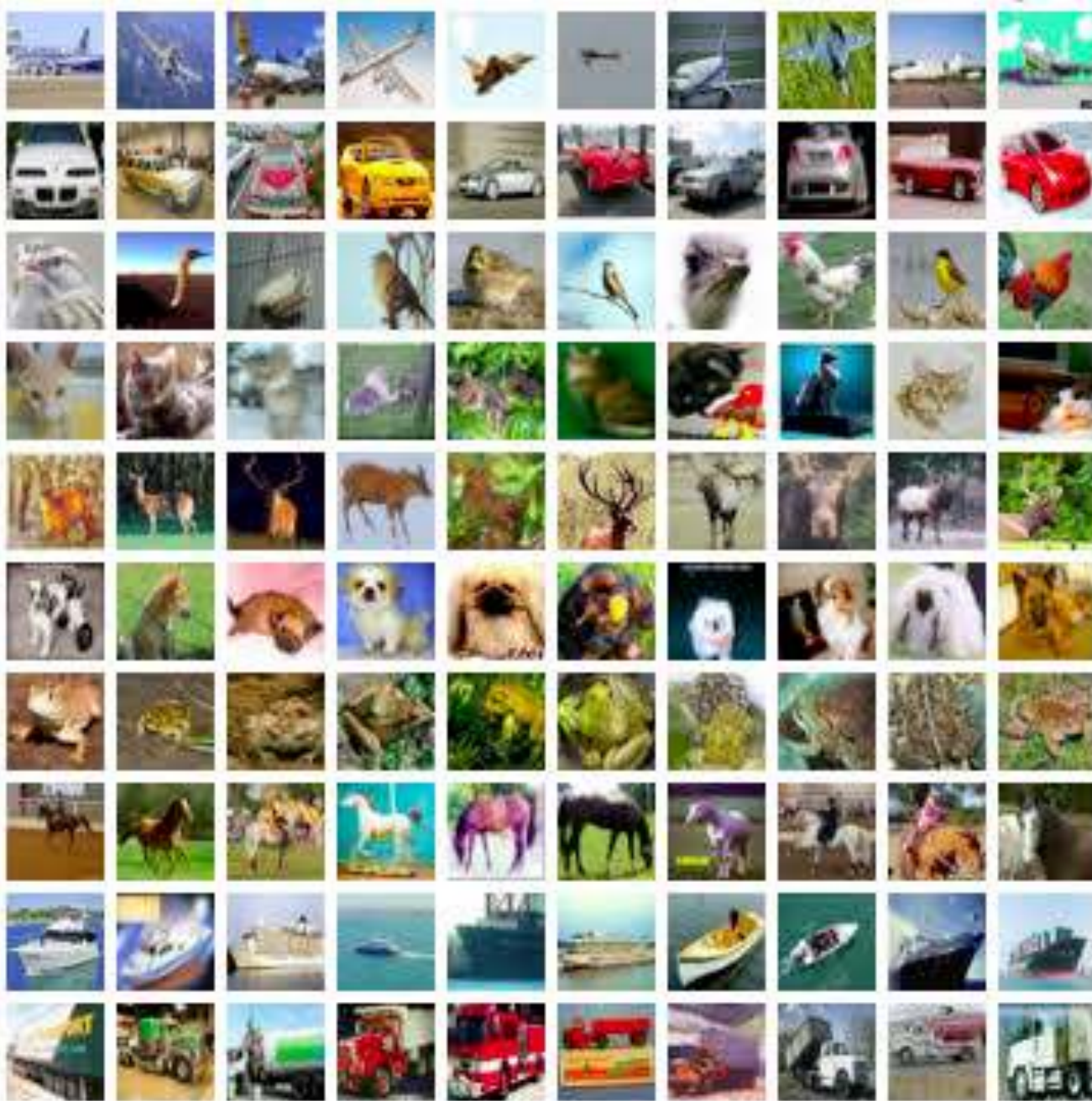
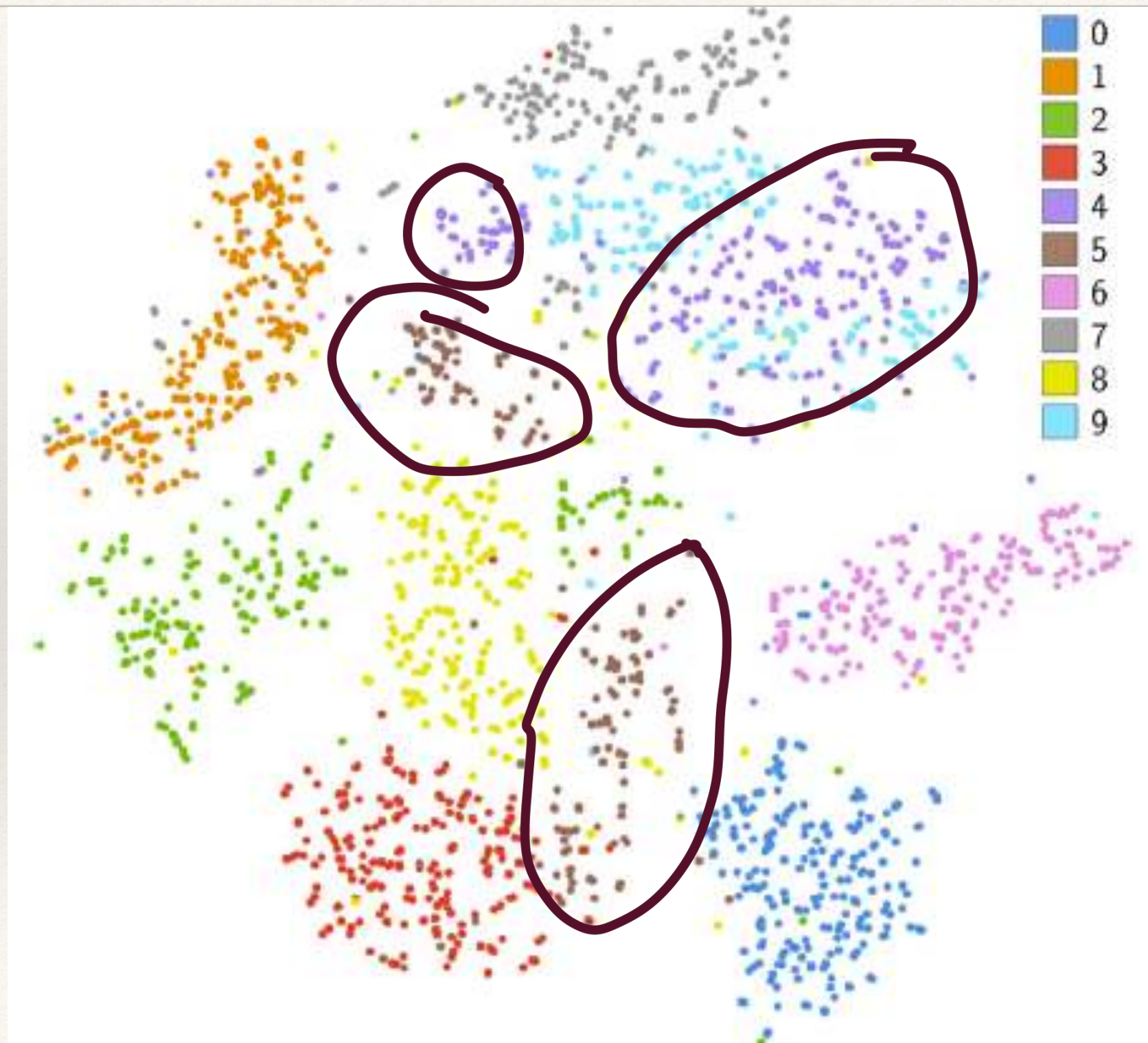


Table 1. Test Set Accuracies for our Two Architectures

Model \ Dataset	MLP	CNN	State-of-the-art
MNIST	98.52%	99.62%	99.79% [47]
SVHN	77.38%	93.76%	98.08% [23]
CIFAR-10	52.91%	79.19%	91.78% [23]

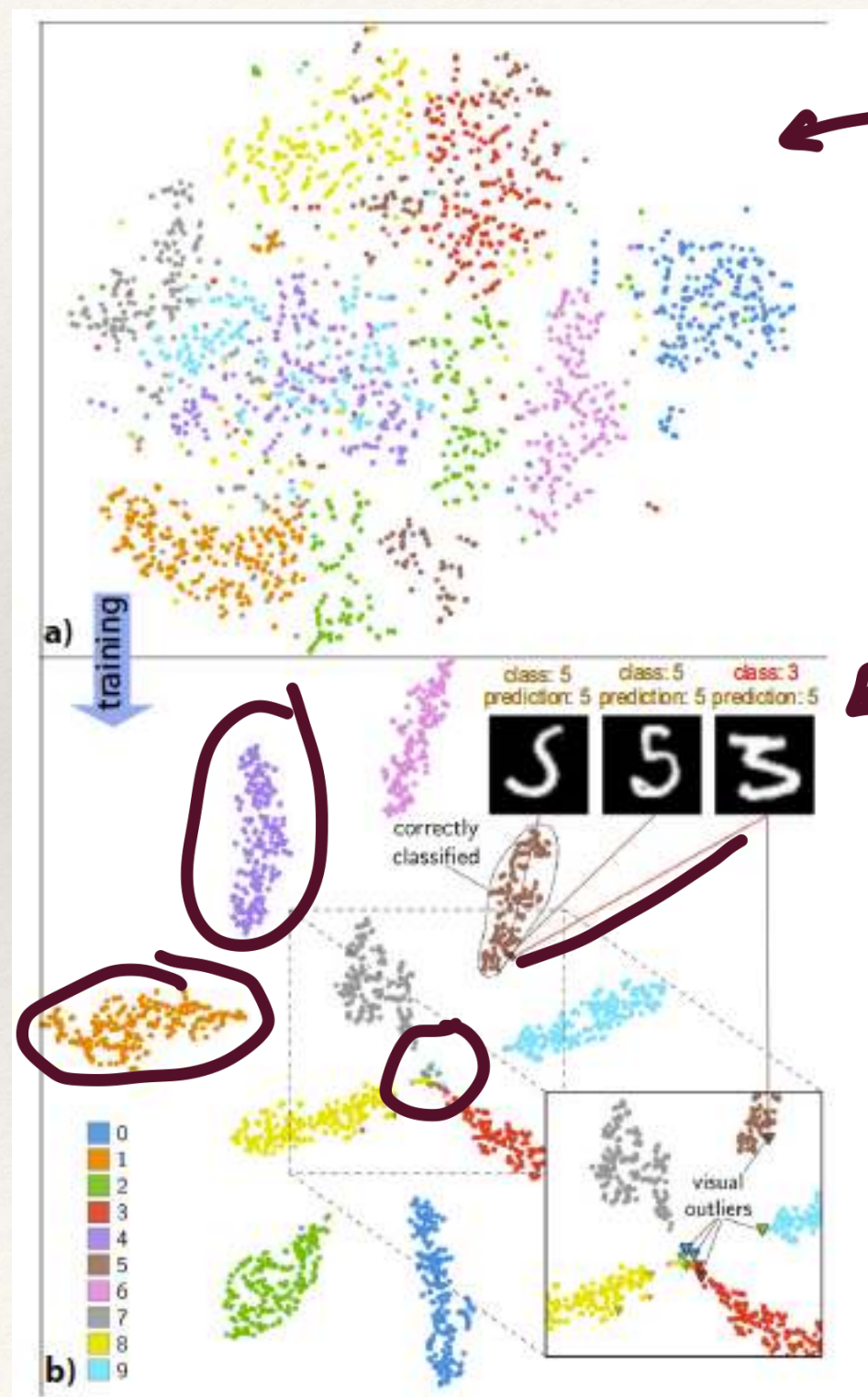
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4 4

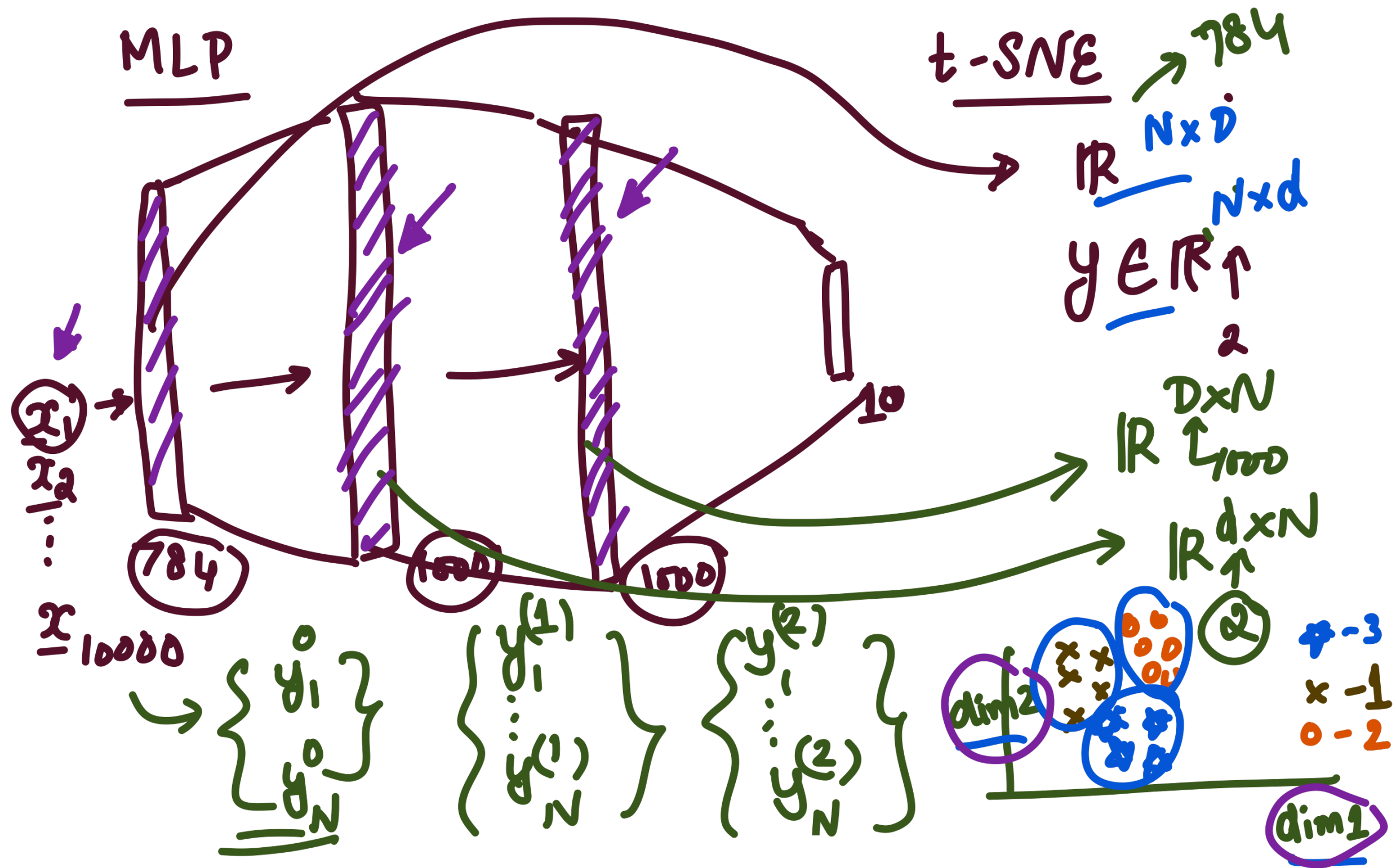
tSNE
projection
of MNIST
Images

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tSNE
projection
of last layer
of the neural network.

Fig. 3. Projection of the last MLP hidden layer activations MNIST test subset. a) Before training (NH: 83.78%). b) After training (NH: 98.36%, AC: 99.15%). Inset shows classification of visual outliers.



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0
1
2
3
4
5
6
7
8
9

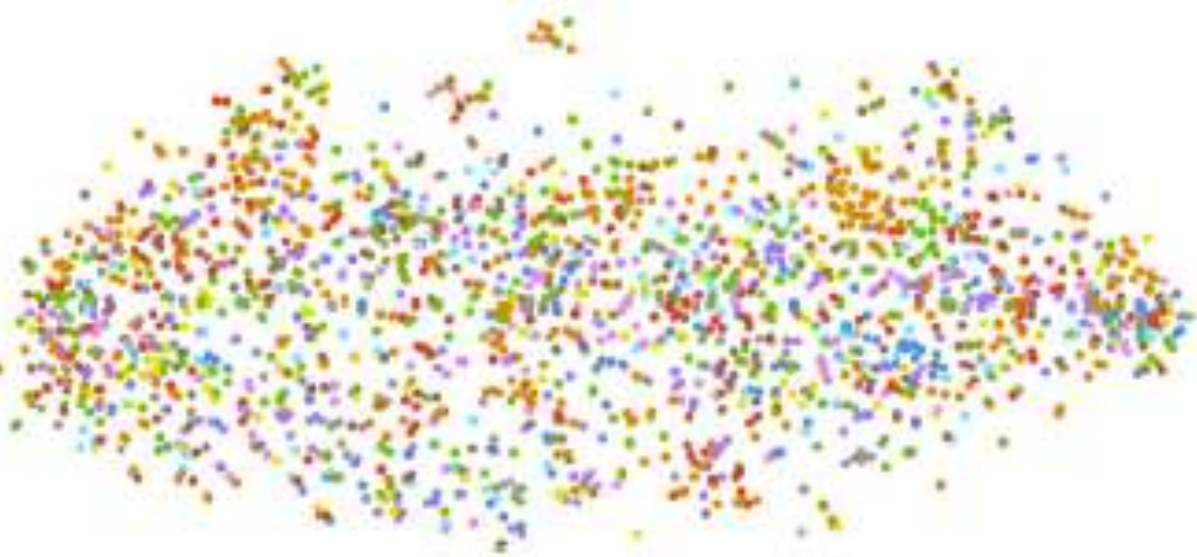


Fig. 4. Projection of the last MLP hidden layer activations before training, SVHN test subset (NH: 20.94%). Poor class separation is visible.

SVHN

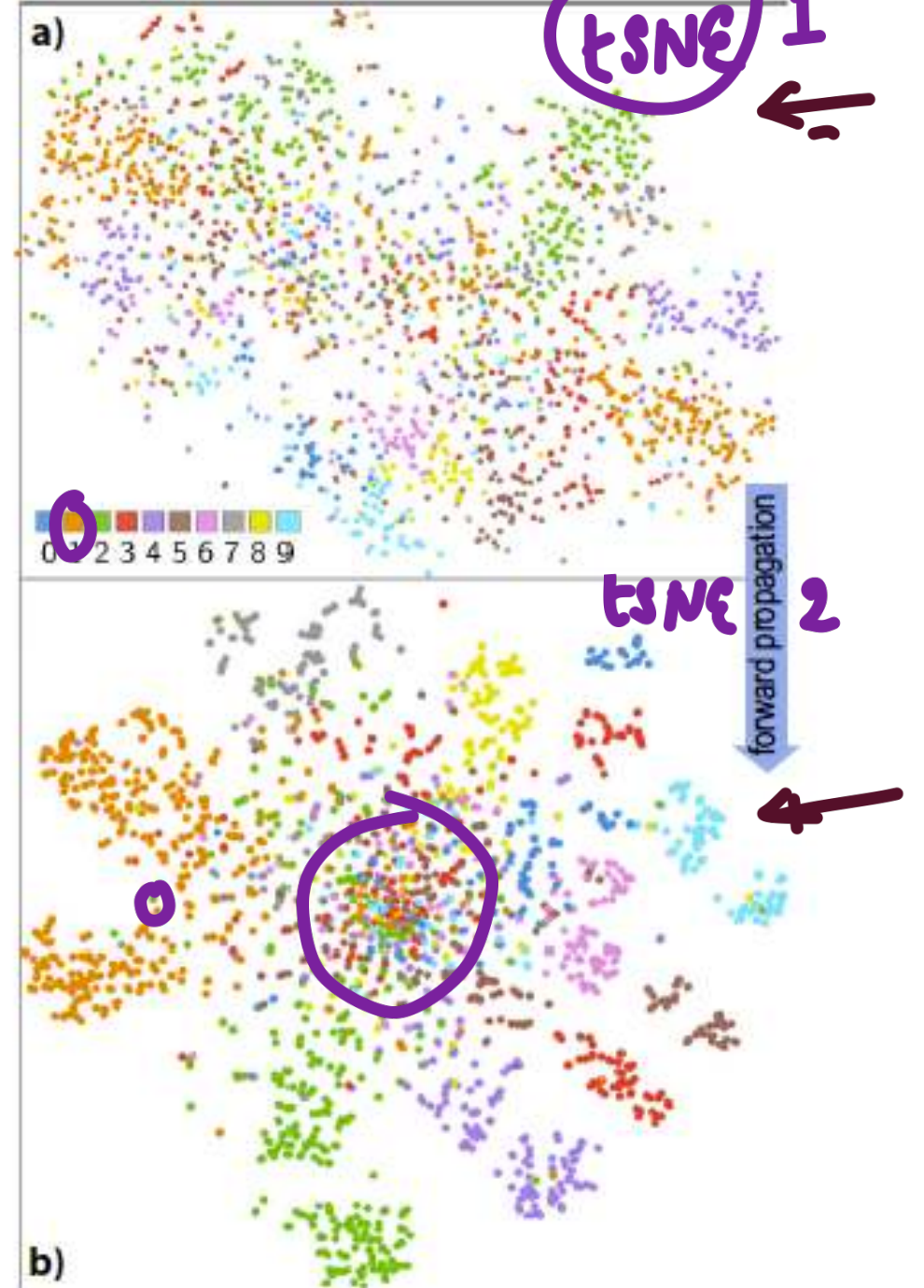


Fig. 5. Projection of the MLP hidden layer activations after training, SVHN test subset. a) First hidden layer (NH: 52.78%). b) Last hidden layer (NH: 67%).

CIFAR-10

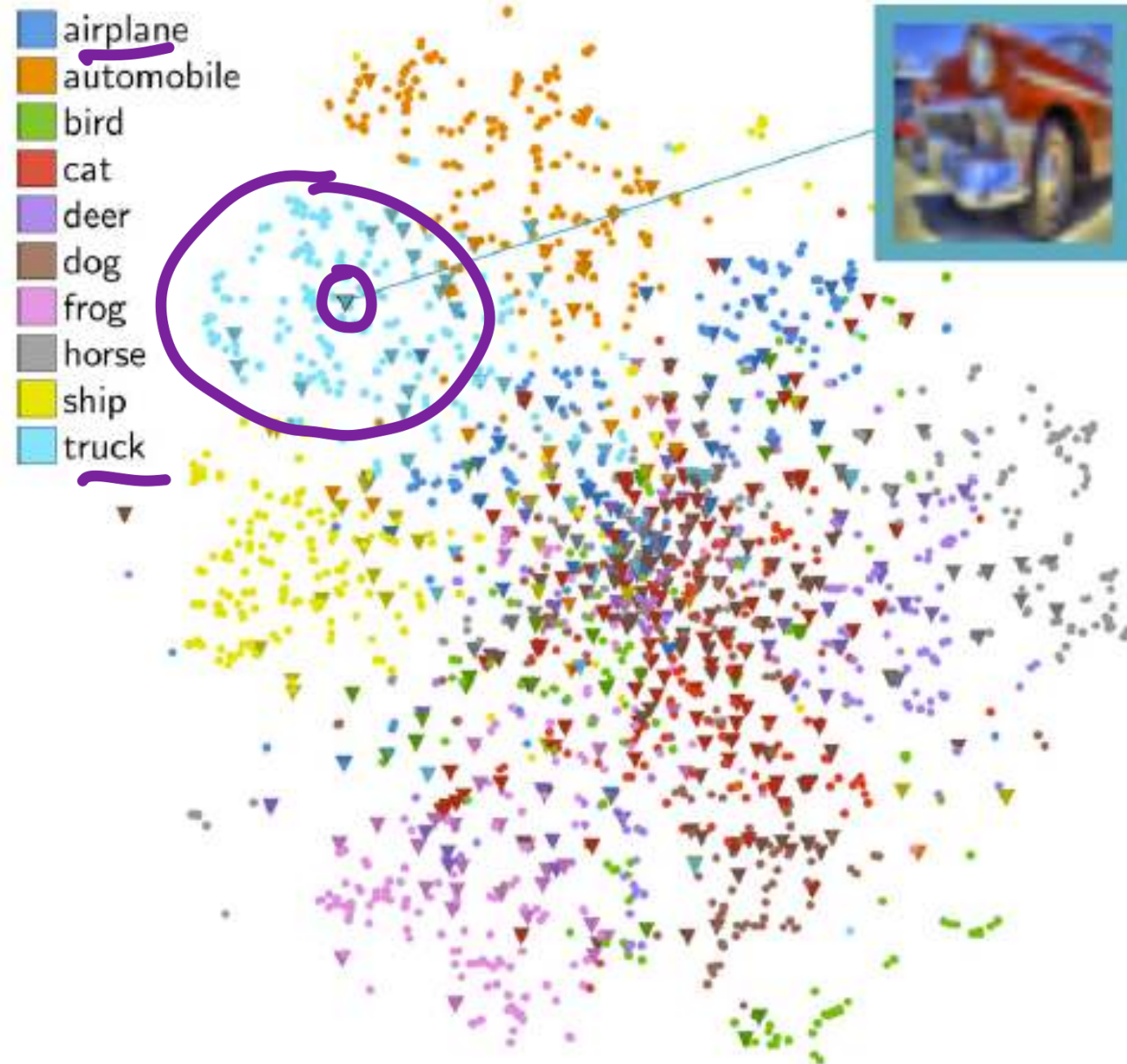


Fig. 9. Projection of last CNN hidden layer activations after training, *CIFAR-10* test subset (NH: 53.43%, AC: 78.7%).

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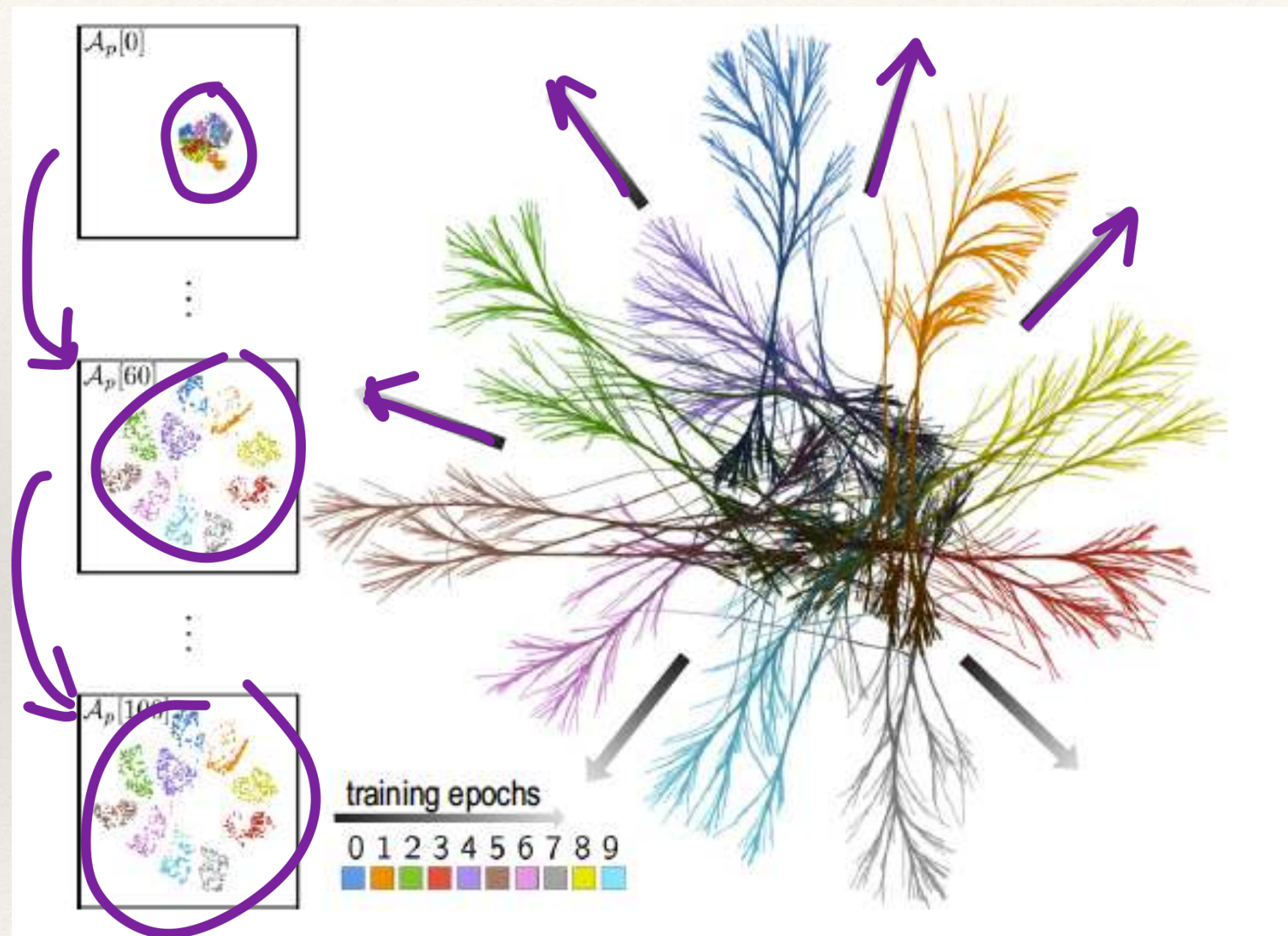


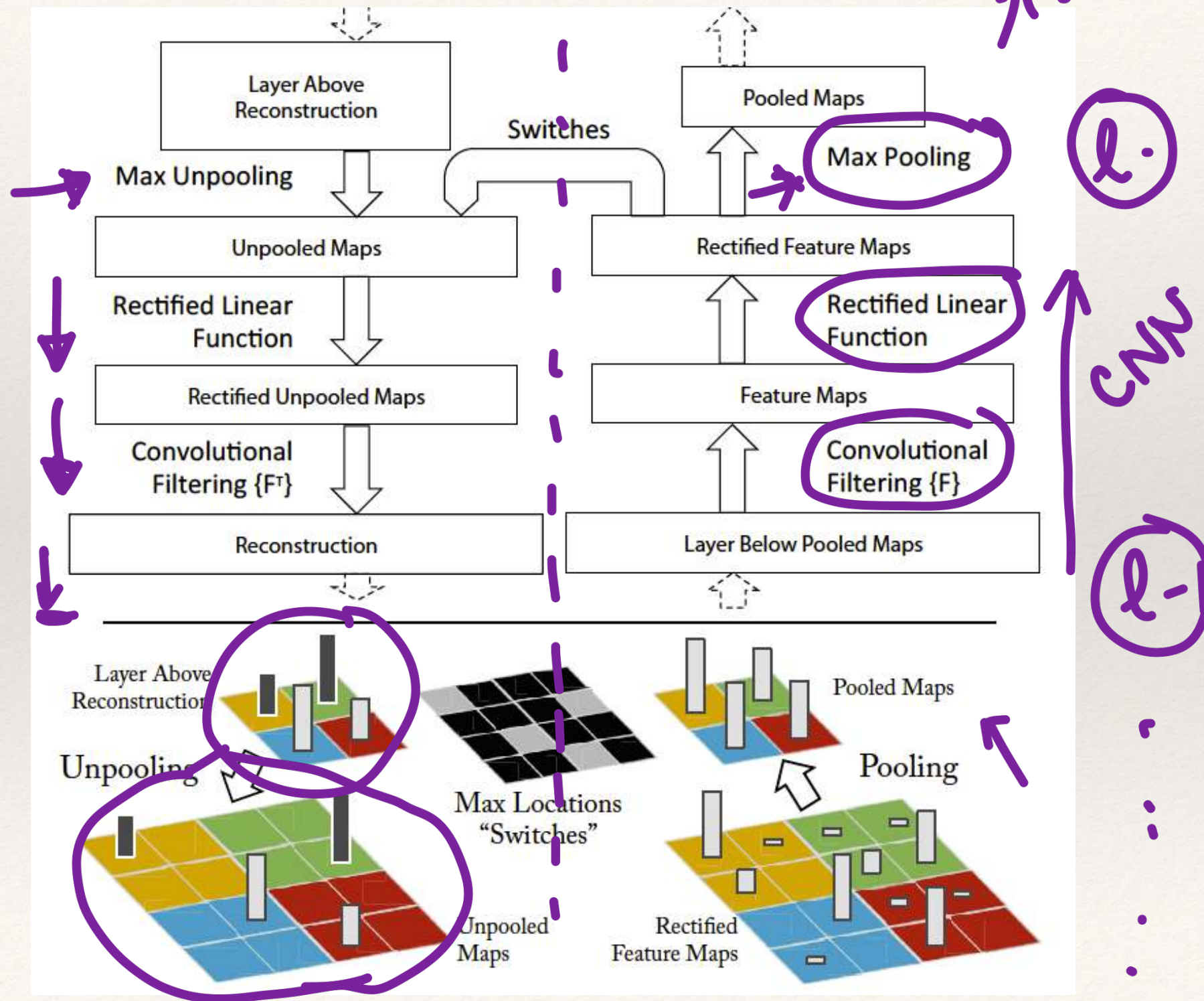
Fig. 11. Inter-epoch evolution, last CNN hidden layer, epochs 0-100, in steps of 20, *MNIST* test subset. Brighter trail parts show later epochs.

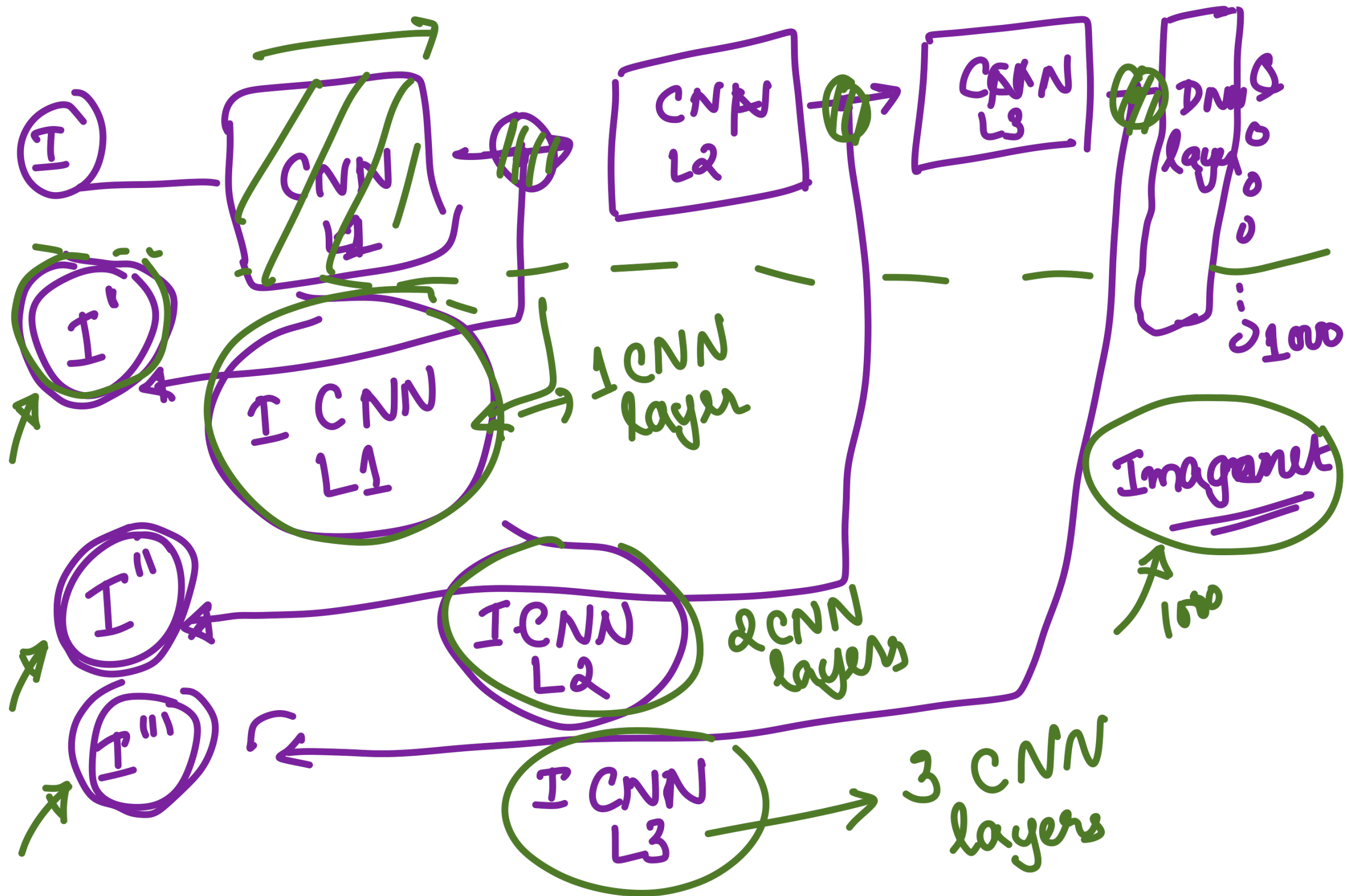
Visualizing and Understanding Convolutional Networks

Matthew D. Zeiler and Rob Fergus

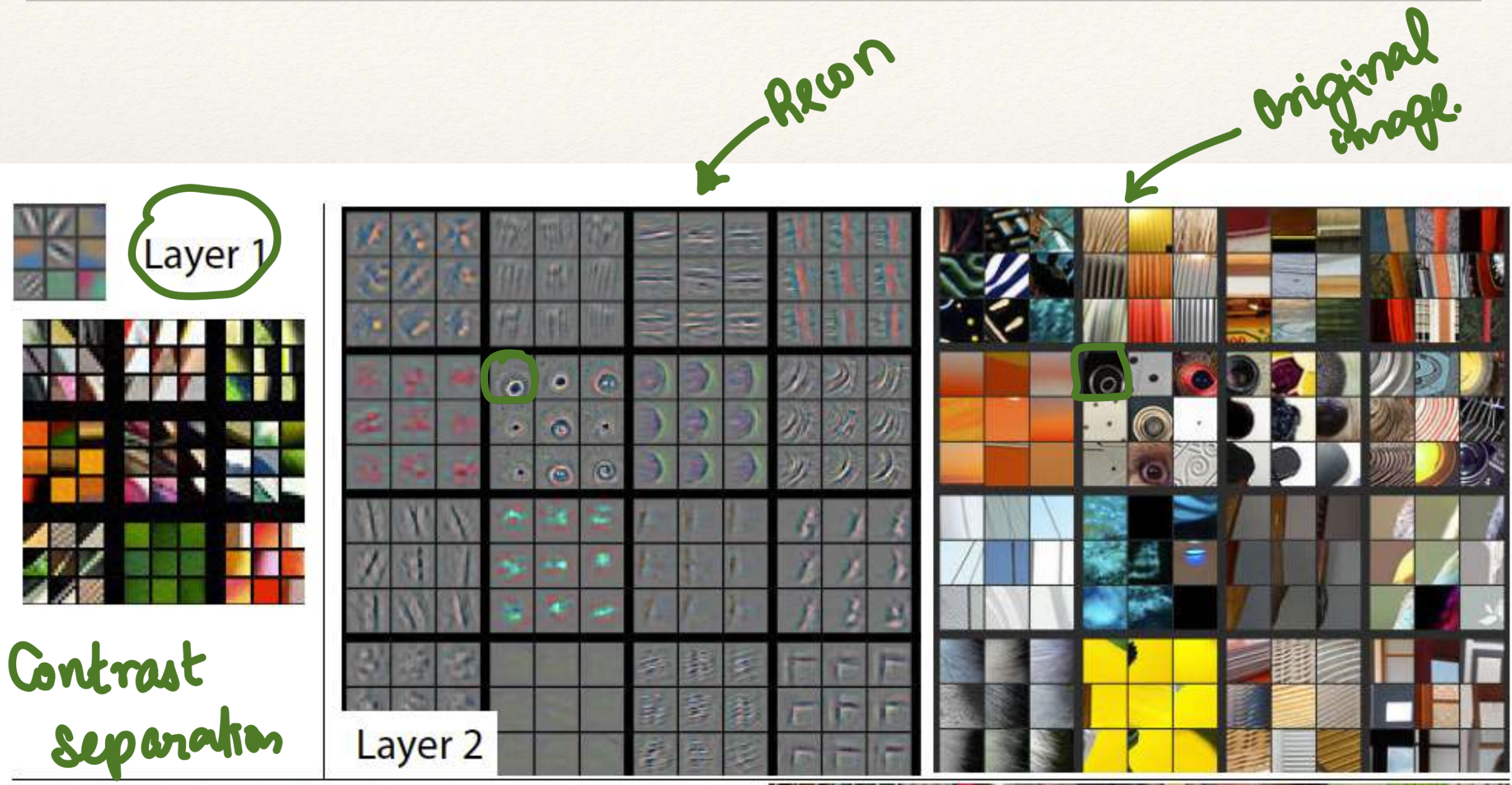
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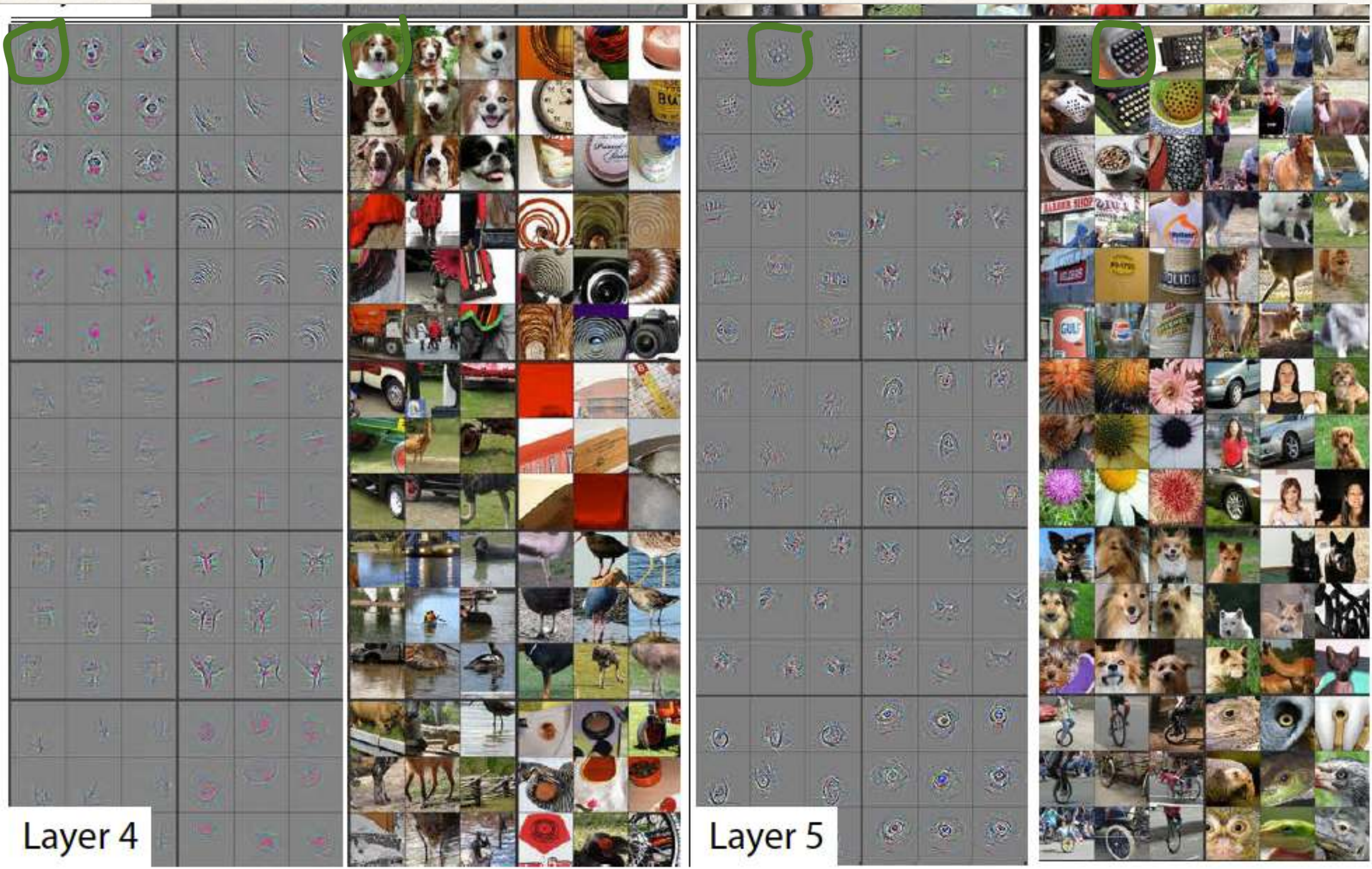




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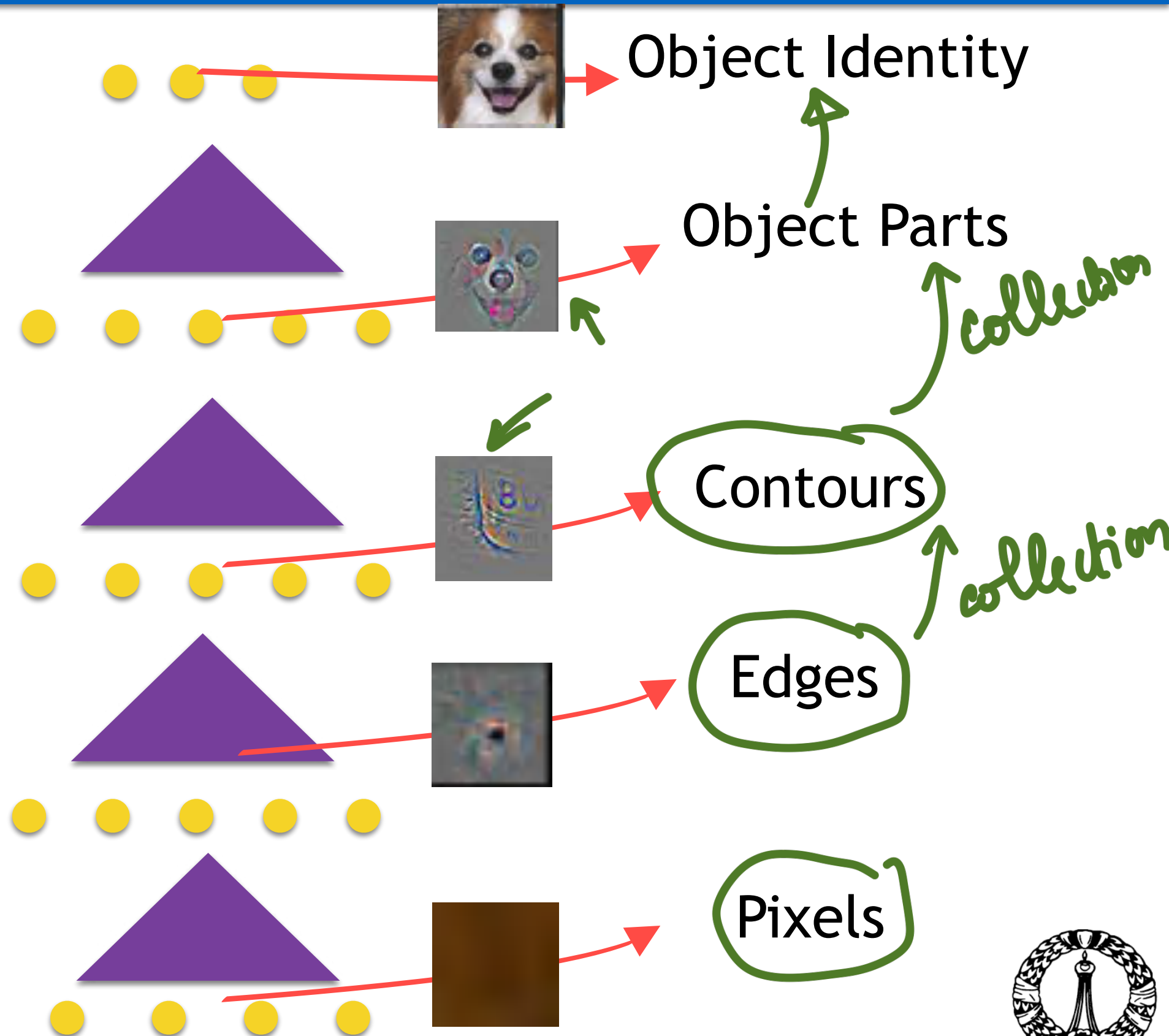


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Representation Learning in Deep Networks

[Zeiler, 2014]



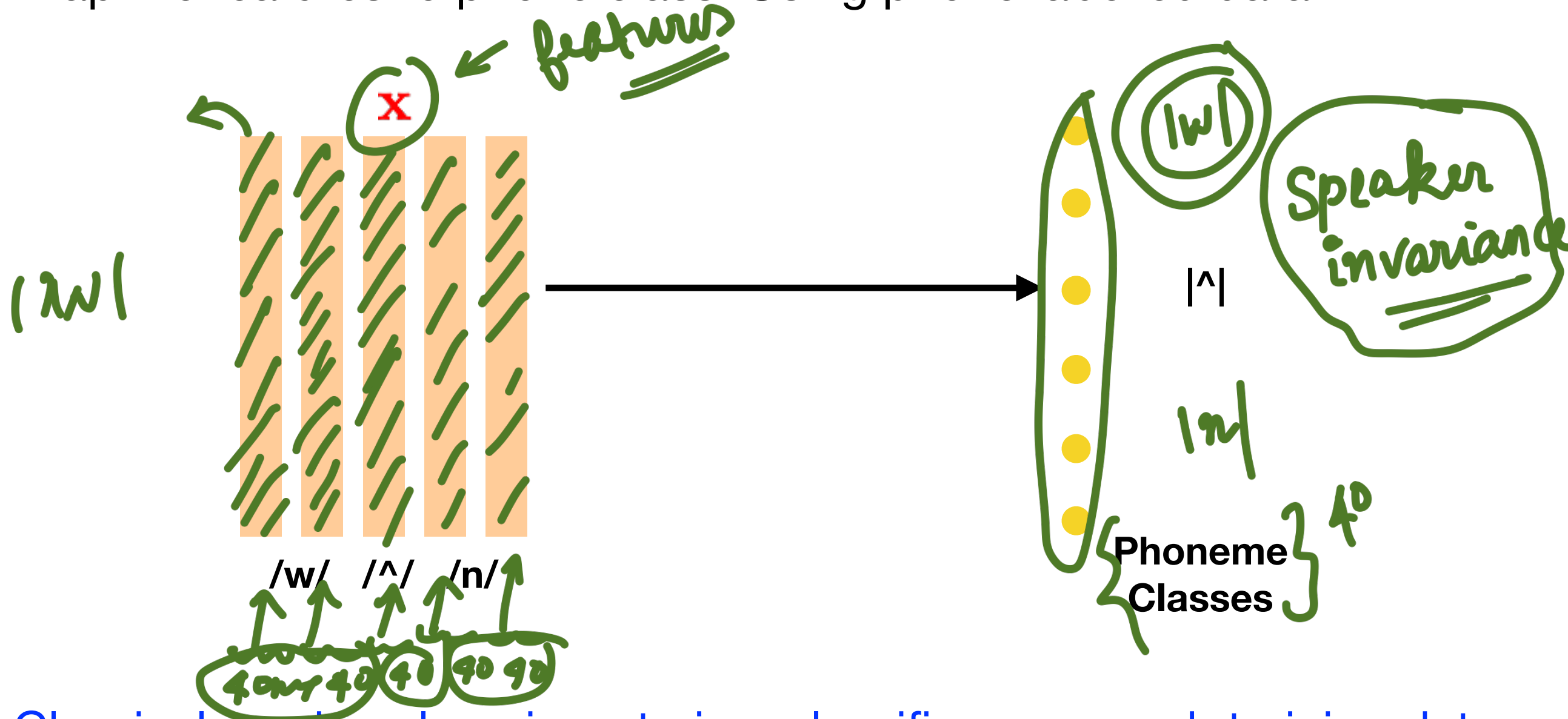
UNDERSTANDING HOW DEEP BELIEF NETWORKS PERFORM ACOUSTIC MODELLING

Garcia-Romero, Daniel, et al. "Speaker diarization using deep neural network embeddings." *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2017.

Department of Computer Science, University of Toronto

Speech Recognition

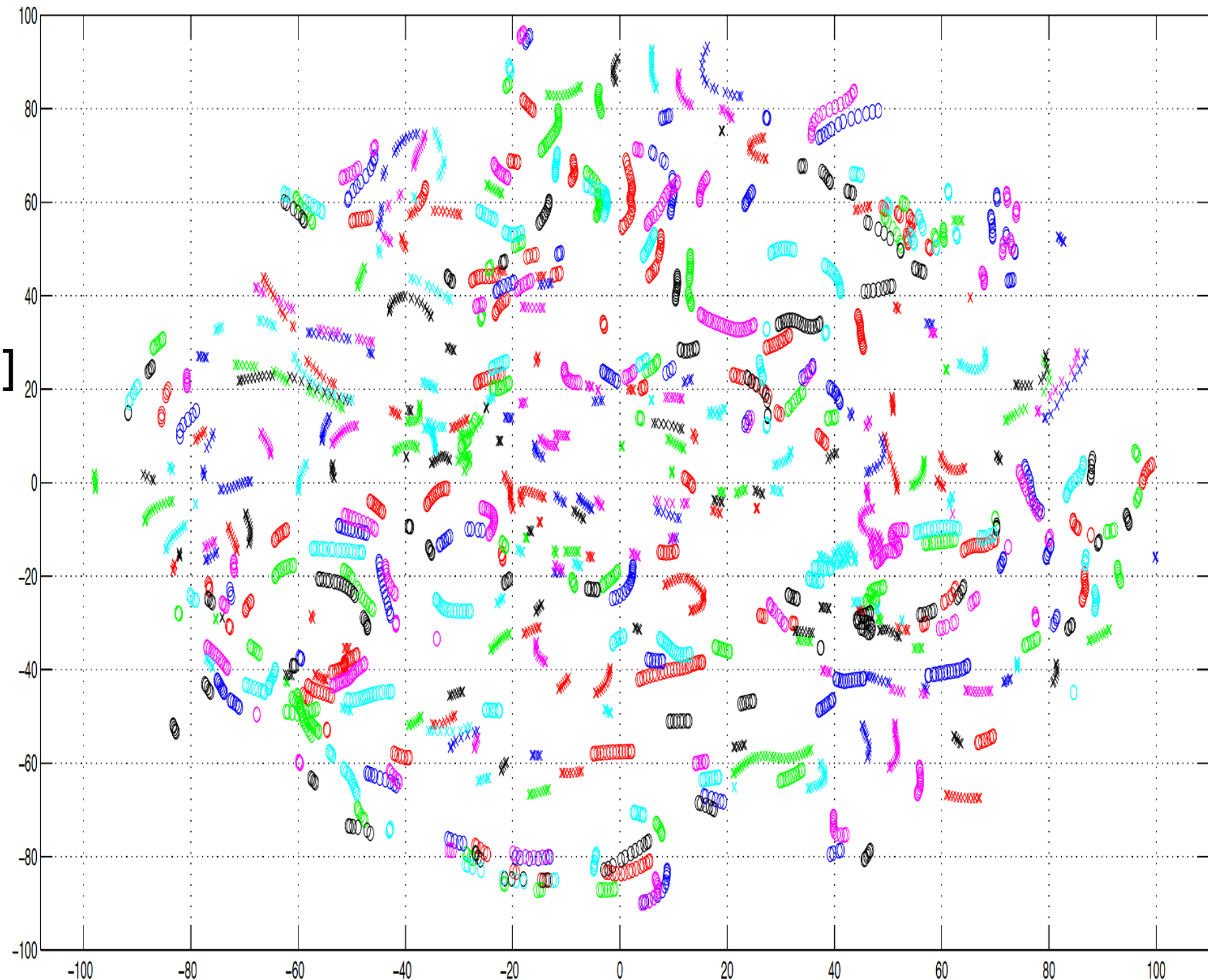
- Map the features to phone class. Using phone labelled data.



- Classical machine learning - train a classifier on speech training data that maps to the target phoneme class.

Understanding DNNs for Speech

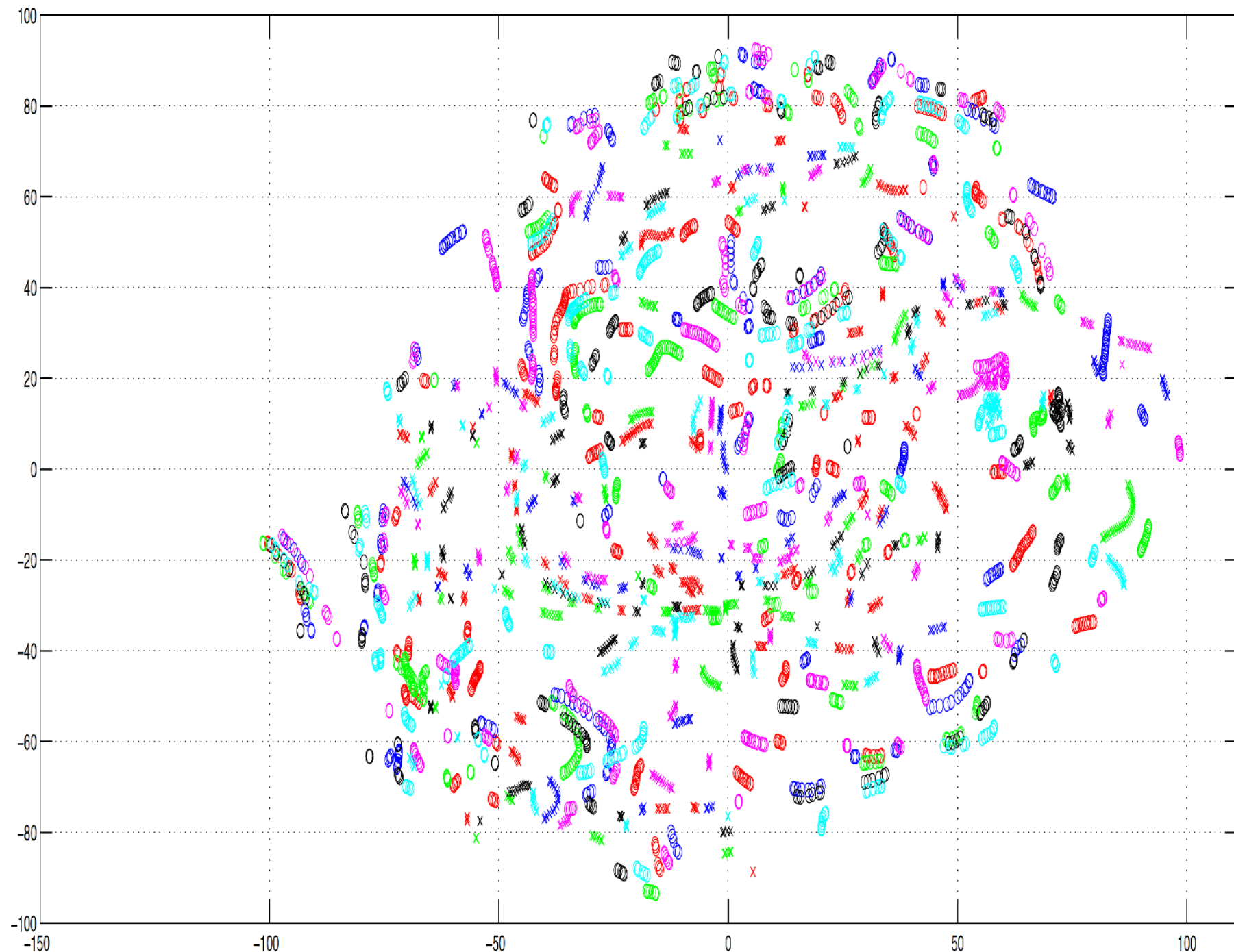
2-D projection of 1st layer DNN



[Abdel Rahman, 2012]

Understanding DNNs for Speech

2-D projection of 2nd layer DNN



[Abdel Rahman, 2012]

Understanding DNNs for Speech

cat drink milk

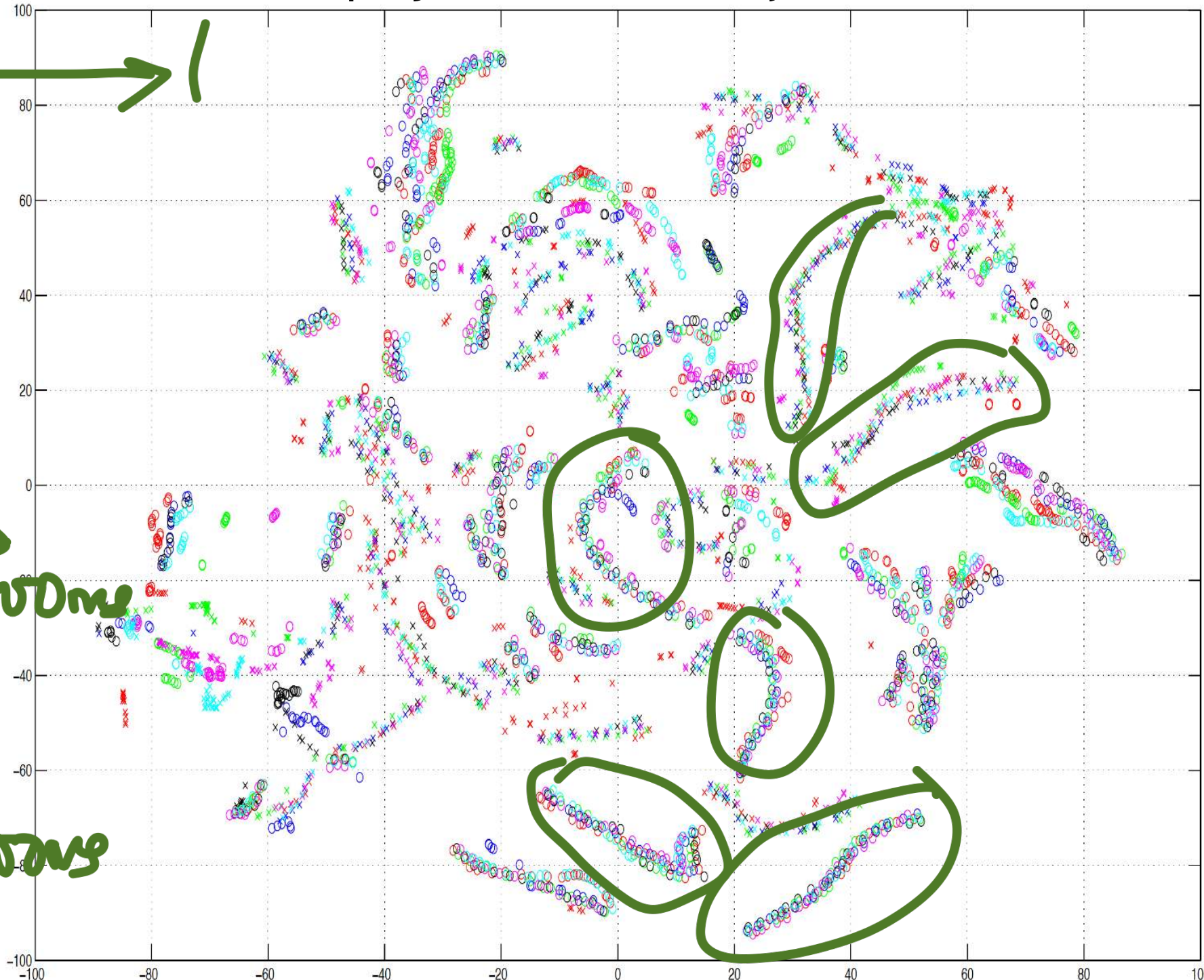
cat drink milk

100

80

60

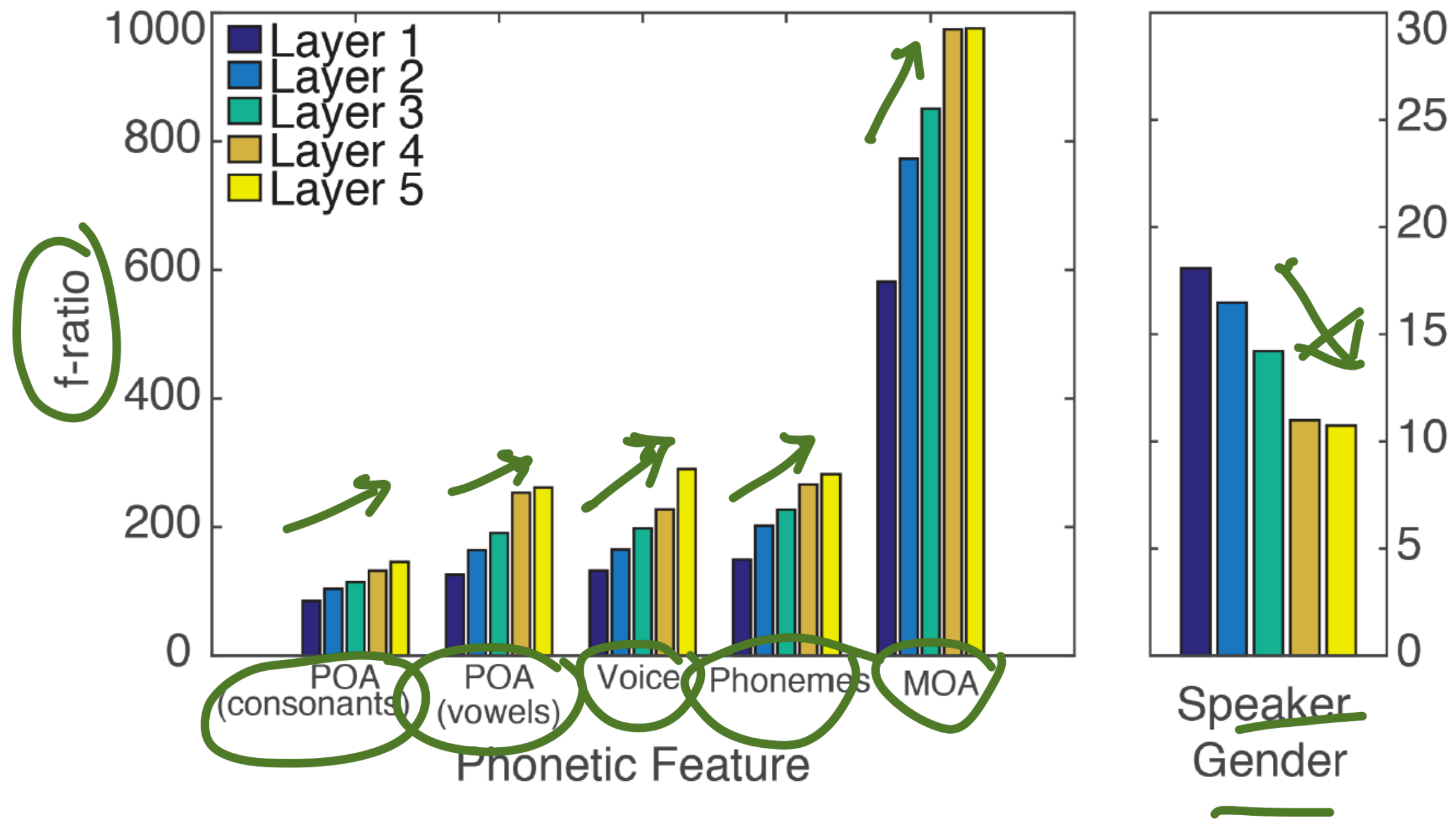
2-D projection of final layer DNN



[Abdel Rahman, 2012]

Hand-drawn diagram illustrating a 2D signal space. The horizontal axis is labeled $|k| |a| (t)$ in green. The vertical axis is labeled $|b| |a| (t)$ in red. A green arrow points right along the horizontal axis, and a red arrow points down along the vertical axis. A green box highlights a region in the upper-left quadrant, and a red box highlights a region in the lower-left quadrant. The diagram is labeled "150ms" and "200ms" on the right side.

Understanding DNNs for Speech



[Nagamine, 2015]